

Felix-LM: Multi-Stream Progressive Merging for Causal Language Modeling

Architecture Design Document & Theoretical Foundation

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Draft — March 2, 2026

Abstract

We propose **Felix-LM**, a novel causal language model architecture derived from the computational principles of the Felix multi-agent framework. Where standard transformer architectures process representations through a uniform stack of identical layers, Felix-LM introduces *Multi-Stream Progressive Merging* (MSPM): the input is processed by multiple independent computational streams that progressively merge into a single output stream through a sequence of structured convergence operations.

We formalize the claim that this architecture induces a *helical funnel trajectory* in representation space—a path characterized by simultaneous rotational mixing and radial compression as representations move through the network’s depth. We provide the mathematical framework for this trajectory, specify the architecture precisely, analyze its gradient flow and information-theoretic properties, and propose an ablation study to test whether the multi-stream progressive merging inductive bias improves perplexity on standard language modeling benchmarks relative to parameter-matched uniform transformers.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	The Felix Framework	3
1.3	Thesis Statement	3
2	Mathematical Preliminaries	4
2.1	Notation	4
2.2	Representation Space and Trajectories	4
2.3	Rotary Position Embeddings (RoPE)	5
3	Architecture	5
3.1	Input Embedding and Stream Initialization	5
3.2	Intra-Stream Computation	5
3.3	Merge Operation	6
3.3.1	Cross-Stream Attention at Merge Points	6
3.3.2	The Merge Projection as Rotation	7
3.4	Depth-Extended Rotary Embeddings	7
3.5	Confidence-Gated Early Exit	8
3.5.1	Cross-Stream Agreement as Confidence	8

3.5.2	Training Objective with Deep Supervision	8
3.6	Output Head	8
3.7	Full Forward Pass	8
4	Theoretical Analysis	9
4.1	Information-Theoretic Capacity	9
4.2	Gradient Flow Analysis	10
4.3	Representational Geometry	10
5	Connection to the Felix Framework	11
6	Proposed Experiments	11
6.1	Ablation Study Design	11
6.2	Hypotheses	12
6.3	Diagnostic Metrics	12
7	Extensions and Open Questions	13
8	Related Work	13
9	Discussion	14
A	Initialization Strategy	15
B	Parameter Count Calculations	15

1 Introduction

1.1 Motivation

The dominant architecture for autoregressive language modeling is the decoder-only transformer [1, 2], consisting of L structurally identical layers, each comprising multi-head self-attention and a position-wise feedforward network. Every layer operates on a representation of fixed dimension d_{model} , and the output of each layer is added to the input via a residual connection. We refer to this as the *uniform* architecture.

Empirical evidence suggests that this uniformity is suboptimal: early layers in trained transformers predominantly capture local syntactic features, while later layers encode global semantic and reasoning structure [3, 4]. The uniform architecture allocates identical computational resources to both regimes, suggesting that a *heterogeneous* architecture—one whose structure varies with depth—might achieve better performance per parameter.

1.2 The Felix Framework

The Felix multi-agent framework¹ provides the conceptual foundation for this work. Felix models a team of AI agents traversing a helical path in 3D space, governed by the parametric equations:

$$x(t) = R(t) \cos(2\pi nt), \quad (1)$$

$$y(t) = R(t) \sin(2\pi nt), \quad (2)$$

$$z(t) = H(1 - t), \quad (3)$$

where $t \in [0, 1]$ is the progression parameter, n is the number of helical turns, H is the total height, and the radius follows an exponential tapering function:

$$R(t) = R_{\min} \left(\frac{R_{\max}}{R_{\min}} \right)^{1-t}. \quad (4)$$

In Felix, agents at the top of the helix ($t \approx 0$) perform broad, divergent exploration; agents at the bottom ($t \approx 1$) perform focused, convergent synthesis. A central hub (the *CentralPost*) sits on the helix’s axis and performs final integration of all agent outputs. The key computational principles are:

- (i) **Parallel exploration, progressive convergence:** Multiple independent processing paths narrow to a single output.
- (ii) **Depth-dependent computation type:** Early processing is broad and divergent; late processing is focused and convergent.
- (iii) **Confidence-gated resource allocation:** Computation is allocated adaptively based on a confidence signal.
- (iv) **Structured communication at convergence points:** Cross-stream information exchange occurs at specific merge events, not globally at every step.

1.3 Thesis Statement

We propose that these four principles can be instantiated as an inductive bias in the architecture of a causal language model, yielding a model that traces a *helical funnel* in representation space: a trajectory characterized by monotonic reduction in the number of independent processing streams (the “funnel”) combined with learned rotational mixing at stream-merge boundaries

¹Developed by C. [Surname]; codebase and documentation available at [repository].

(the “helix”). We hypothesize that this inductive bias improves language modeling perplexity relative to a parameter-matched uniform transformer, particularly on sequences requiring multi-step reasoning.

2 Mathematical Preliminaries

2.1 Notation

Let $\mathcal{V}$ denote a vocabulary of size V . An input sequence is $\mathbf{x} = (x_1, \dots, x_T)$ with $x_i \in \mathcal{V}$. A causal language model defines $p(x_{t+1} \mid x_1, \dots, x_t)$ for all t .

We write $\mathbb{R}^d$ for d -dimensional Euclidean space. For a matrix $W \in \mathbb{R}^{m \times n}$, we write $W^\top$ for its transpose and $\|W\|_F$ for its Frobenius norm. The identity matrix is $I_d \in \mathbb{R}^{d \times d}$. We write $\sigma(\cdot)$ for the logistic sigmoid function and $\text{softmax}(\cdot)$ for the softmax function.

2.2 Representation Space and Trajectories

Definition 2.1 (Representation trajectory). Let $f : \mathbb{R}^{d_{\text{in}}} \rightarrow \mathbb{R}^{d_{\text{out}}}$ be a neural network of depth L with intermediate representations $\mathbf{h}^{(0)}, \mathbf{h}^{(1)}, \dots, \mathbf{h}^{(L)}$. The *representation trajectory* of an input $\mathbf{x}$ is the sequence

$$\gamma(\mathbf{x}) = (\mathbf{h}^{(0)}, \mathbf{h}^{(1)}, \dots, \mathbf{h}^{(L)}),$$

where $\mathbf{h}^{(\ell)} \in \mathbb{R}^{d_\ell}$ is the representation at depth ℓ . When d_ℓ varies with ℓ , we embed all representations into a common ambient space $\mathbb{R}^{d_{\text{max}}}$ via zero-padding for the purpose of geometric analysis.

Definition 2.2 (Effective dimensionality). For a representation $\mathbf{h} \in \mathbb{R}^d$, the *effective dimensionality* with respect to a gate vector $\mathbf{g} \in [0, 1]^d$ is

$$d_{\text{eff}}(\mathbf{h}, \mathbf{g}) = \|\mathbf{g}\|_1 = \sum_{i=1}^d g_i.$$

This generalizes hard dimensionality reduction: when $g_i \in \{0, 1\}$, d_{eff} counts the number of active dimensions. For soft gates $g_i \in (0, 1)$, it provides a smooth interpolation.

Definition 2.3 (Helical trajectory in representation space). A representation trajectory γ is *helical* if there exists a sequence of orthogonal transformations $\{Q^{(\ell)} \in O(d_{\text{max}})\}$ and a monotonically decreasing effective dimensionality sequence $\{d_{\text{eff}}^{(\ell)}\}$ such that the transition between successive representations can be decomposed as:

$$\mathbf{h}^{(\ell+1)} = Q^{(\ell)} \cdot (\mathbf{g}^{(\ell)} \odot f^{(\ell)}(\mathbf{h}^{(\ell)})), \quad (5)$$

where $f^{(\ell)}$ is the layer computation, $\mathbf{g}^{(\ell)}$ is a gating vector with $\|\mathbf{g}^{(\ell)}\|_1 \leq \|\mathbf{g}^{(\ell-1)}\|_1$, $\odot$ denotes elementwise multiplication, and $Q^{(\ell)}$ is an orthogonal rotation.

The *angular velocity* of the trajectory is $\omega^{(\ell)} = \arccos(\frac{\text{tr}(Q^{(\ell)})}{d_{\text{max}}})$, and the *radial compression rate* is $\rho^{(\ell)} = d_{\text{eff}}^{(\ell+1)} / d_{\text{eff}}^{(\ell)} \leq 1$.

Remark 2.4. In a standard uniform transformer with residual connections, the trajectory is approximately *linear* (representations are incrementally shifted in a consistent direction) with constant effective dimensionality. The helical trajectory introduces both rotation and compression, forcing the network to progressively refine and consolidate its representation. This is the mathematical content of the Felix helical metaphor.

2.3 Rotary Position Embeddings (RoPE)

Rotary Position Embeddings [5] encode token position m by applying a block-diagonal rotation matrix $\text{Rot}(m) \in \mathbb{R}^{d \times d}$ to query and key vectors before computing attention. For dimension pair $(2i, 2i + 1)$ with frequency $\theta_i = 10000^{-2i/d}$:

$$\text{Rot}(m)_{2i:2i+1} = \begin{pmatrix} \cos(m\theta_i) & -\sin(m\theta_i) \\ \sin(m\theta_i) & \cos(m\theta_i) \end{pmatrix}. \quad (6)$$

The key property is that the inner product between rotated queries and keys depends only on relative position: $\langle \text{Rot}(m)\mathbf{q}, \text{Rot}(n)\mathbf{k} \rangle = \langle \text{Rot}(m - n)\mathbf{q}, \mathbf{k} \rangle$.

3 Architecture

Felix-LM is organized into K *stages*, indexed $k = 0, 1, \dots, K-1$. Stage k contains S_k independent processing *streams*, each of model dimension d_k . The number of streams decreases monotonically: $S_0 > S_1 > \dots > S_{K-1} = 1$. Between consecutive stages, a *merge operation* reduces the stream count by combining pairs of streams. Each stream within a stage consists of L_k transformer layers (self-attention + FFN), with architecture details that may vary by stage.

The overall computation proceeds:

$$\underbrace{\text{Embed}}_{\text{input}} \rightarrow \underbrace{S_0 \text{ streams} \times L_0 \text{ layers}}_{\text{Stage 0}} \xrightarrow{\text{Merge}} \underbrace{S_1 \text{ streams} \times L_1 \text{ layers}}_{\text{Stage 1}} \xrightarrow{\text{Merge}} \dots \xrightarrow{\text{Merge}} \underbrace{1 \text{ stream} \times L_{K-1} \text{ layers}}_{\text{Stage } K-1} \quad (7)$$

3.1 Input Embedding and Stream Initialization

Given an input sequence $\mathbf{x} = (x_1, \dots, x_T)$, we compute a shared token embedding:

$$\mathbf{e}_t = \text{Embed}(x_t) \in \mathbb{R}^{d_{\text{embed}}}, \quad t = 1, \dots, T. \quad (8)$$

Each stream $s \in \{1, \dots, S_0\}$ in Stage 0 receives the input through a learned *stream projection*:

$$\mathbf{h}_{t,s}^{(0,0)} = W_s^{\text{init}} \mathbf{e}_t + \mathbf{b}_s^{\text{init}}, \quad W_s^{\text{init}} \in \mathbb{R}^{d_0 \times d_{\text{embed}}}, \quad (9)$$

where each projection W_s^{init} is initialized to encourage diverse initial representations across streams (see §A).

Remark 3.1. Each stream can be understood as attending to a different “aspect” of the input, analogous to how Felix spawns agents of different types (Research, Analysis, Critic) that examine the same task from different perspectives. The stream projections play the role of role specialization.

3.2 Intra-Stream Computation

Within stage k , each stream s independently processes its representation through L_k transformer layers. We denote the representation at layer ℓ within stage k , stream s , token position t as $\mathbf{h}_{t,s}^{(k,\ell)} \in \mathbb{R}^{d_k}$.

Each layer consists of:

$$\tilde{\mathbf{h}} = \mathbf{h}_{t,s}^{(k,\ell)} + \text{Attn}_{k,\ell,s}(\text{LN}(\mathbf{h}_{:,s}^{(k,\ell)}))_t, \quad (10)$$

$$\mathbf{h}_{t,s}^{(k,\ell+1)} = \tilde{\mathbf{h}} + \text{FFN}_{k,\ell,s}(\text{LN}(\tilde{\mathbf{h}})), \quad (11)$$

where LN denotes RMSNorm [6], and the attention and FFN modules are specific to stage k , layer ℓ , and stream s . The notation $\mathbf{h}_{:,s}^{(k,\ell)}$ denotes the full sequence of representations for stream s (used as keys/values in attention).

Heterogeneous attention by stage. A key design principle is that the *type* of attention varies with stage, reflecting Felix’s principle of depth-dependent computation:

Stage	Felix Analog	Attention Type	Rationale
$k = 0$	Research (exploration)	Linear / SSM	$\mathcal{O}(T)$: broad, cheap context
$k = 1, \dots, K-2$	Analysis (convergence)	Sliding window	$\mathcal{O}(Tw)$: local precision
$k = K-1$	CentralPost (synthesis)	Full causal	$\mathcal{O}(T^2)$: maximum precision

This allocation concentrates quadratic-cost attention at the final stage, where only a single stream operates, and uses efficient linear-time mechanisms in the early stages where multiple parallel streams would otherwise multiply the cost.

Weight sharing. Streams within the same stage may share weights (all streams identical, differentiated only by their initial projection) or have independent weights (each stream is a distinct sub-network). We denote these configurations as SHARED and INDEPENDENT, respectively, and treat this as an ablation variable.

3.3 Merge Operation

The merge operation is the architectural core of Felix-LM, implementing the progressive convergence that defines the helical funnel.

Definition 3.2 (Stream merge). Let streams s and s' at the end of stage k have representations $\mathbf{h}_{t,s}^{(k,L_k)}, \mathbf{h}_{t,s'}^{(k,L_k)} \in \mathbb{R}^{d_k}$ for each token t . The *merge operation* produces a single stream representation $\mathbf{m}_t \in \mathbb{R}^{d_{k+1}}$ for the next stage:

$$\mathbf{m}_t = \underbrace{P_k}_{\text{projection}} \left(\underbrace{\mathbf{g}_k([\mathbf{h}_{t,s}; \mathbf{h}_{t,s'}])}_{\text{gated combination}} \right), \quad (12)$$

where $[\cdot; \cdot]$ denotes concatenation along the feature axis, $P_k \in \mathbb{R}^{d_{k+1} \times 2d_k}$ is a learned projection, and $\mathbf{g}_k: \mathbb{R}^{2d_k} \rightarrow [0, 1]^{2d_k}$ is a gating function:

$$\mathbf{g}_k(\mathbf{z}) = \sigma(W_k^g \mathbf{z} + \mathbf{b}_k^g), \quad W_k^g \in \mathbb{R}^{2d_k \times 2d_k}. \quad (13)$$

The full merge operation is thus:

$$\mathbf{m}_t = P_k \left(\sigma(W_k^g [\mathbf{h}_{t,s}; \mathbf{h}_{t,s'}] + \mathbf{b}_k^g) \odot [\mathbf{h}_{t,s}; \mathbf{h}_{t,s'}] \right). \quad (14)$$

Remark 3.3 (Connection to Felix’s CentralPost). In Felix, the CentralPost synthesizes agent outputs with adaptive parameters (temperature $\in [0.2, 0.4]$, token budget $\in [1500, 3000]$). The merge gate $\mathbf{g}_k$ plays an analogous role: it learns which features from each stream to preserve (high gate value) versus suppress (low gate value) during synthesis. The projection P_k corresponds to the CentralPost’s final synthesis prompt.

3.3.1 Cross-Stream Attention at Merge Points

Before applying the gated projection, we optionally perform a single layer of *cross-stream attention*, where each stream attends to the other:

$$\hat{\mathbf{h}}_{t,s} = \mathbf{h}_{t,s}^{(k,L_k)} + \text{CrossAttn}(\mathbf{h}_{t,s}^{(k,L_k)}, \mathbf{h}_{t,s'}^{(k,L_k)}), \quad (15)$$

where $\text{CrossAttn}(Q\text{-source}, KV\text{-source})$ is standard multi-head attention with queries from one stream and keys/values from the other. This allows streams to exchange information before fusion, implementing Felix’s principle of structured communication at convergence points. We treat the inclusion of cross-stream attention as an ablation variable.

3.3.2 The Merge Projection as Rotation

We can decompose the merge projection P_k via polar decomposition:

$$P_k = U_k \Sigma_k V_k^\top, \quad (16)$$

where $U_k \in \mathbb{R}^{d_{k+1} \times d_{k+1}}$ and $V_k \in \mathbb{R}^{2d_k \times 2d_k}$ are orthogonal, and Σ_k is the singular value matrix. The matrix $V_k^\top$ rotates the concatenated representation in $\mathbb{R}^{2d_k}$, Σ_k scales (compresses) dimensions, and U_k rotates the result in $\mathbb{R}^{d_{k+1}}$.

This decomposition reveals that every merge naturally performs *rotation followed by compression followed by rotation*—the mathematical structure of a helical step. We may optionally constrain P_k to have orthogonal rows ($P_k P_k^\top = I_{d_{k+1}}$) to guarantee *isometric* compression: the merge preserves inner products and thus similarity structure, ensuring that information is consolidated without distortion. This constraint can be enforced via Cayley parameterization:

$$P_k = (I - A_k)(I + A_k)^{-1} \cdot \Pi, \quad (17)$$

where A_k is a learnable skew-symmetric matrix and $\Pi \in \mathbb{R}^{d_{k+1} \times 2d_k}$ is a fixed truncation matrix. We treat this orthogonality constraint as an ablation variable.

3.4 Depth-Extended Rotary Embeddings

Standard RoPE (§2.3) encodes token position via rotation. We extend this to also encode *depth* (layer index), providing the “twist” of the helical trajectory.

Definition 3.4 (Depth-extended RoPE). At global layer index ℓ_{global} (counting across all stages) and token position m , the rotation applied to dimension pair $(2i, 2i + 1)$ of queries and keys is:

$$\text{Rot}(m, \ell_{\text{global}})_{2i:2i+1} = \begin{pmatrix} \cos \phi_i & -\sin \phi_i \\ \sin \phi_i & \cos \phi_i \end{pmatrix}, \quad (18)$$

where the composite angle is:

$$\phi_i(m, \ell_{\text{global}}) = m \cdot \theta_i^{\text{pos}} + \ell_{\text{global}} \cdot \theta_i^{\text{depth}}, \quad (19)$$

with position frequencies $\theta_i^{\text{pos}} = \beta_{\text{pos}}^{-2i/d}$ and depth frequencies $\theta_i^{\text{depth}} = \alpha_i \cdot \frac{2\pi n}{L_{\text{total}}}$.

Here n is the number of helical turns (directly from the Felix configuration), $L_{\text{total}} = \sum_k L_k$ is the total layer count, β_{pos} is the RoPE base frequency (typically 10000), and α_i is a per-dimension scaling factor that controls the coupling strength between depth and position. Setting $\alpha_i = 0$ recovers standard RoPE; setting $\alpha_i = 1$ applies the full helical twist.

Proposition 3.5 (Relative depth encoding). *Under depth-extended RoPE, the attention score between query at position m , depth ℓ and key at position m' , depth ℓ' in dimension pair i depends only on the relative position $\Delta m = m - m'$ and relative depth $\Delta \ell = \ell - \ell'$:*

$$\langle \text{Rot}(m, \ell) \mathbf{q}_i, \text{Rot}(m', \ell') \mathbf{k}_i \rangle = \langle \text{Rot}(\Delta m, \Delta \ell) \mathbf{q}_i, \mathbf{k}_i \rangle. \quad (20)$$

Proof. This follows from the group homomorphism property of rotation matrices: $\text{Rot}(\phi_1) \text{Rot}(\phi_2) = \text{Rot}(\phi_1 + \phi_2)$, since ϕ_i is linear in both m and ℓ_{global} . $\square$

Remark 3.6. In causal (decoder-only) language modeling, cross-depth attention does not occur—each layer only attends within itself. The depth-extended RoPE nonetheless affects the geometry: it ensures that representations at successive depths exist in *rotated* coordinate frames, which influences how the residual stream accumulates information and how the merge projections interact with different depths’ representations. The relative-depth property becomes directly relevant if skip connections across stages are introduced (§7).

3.5 Confidence-Gated Early Exit

Felix agents with high confidence accelerate through the helix. The LLM analog is that tokens whose next-token prediction is “easy” need not traverse all K stages.

At each merge boundary (after stage k , $k < K - 1$), we attach a lightweight *exit head*:

$$\hat{p}_k(x_{t+1} \mid \mathbf{x}_{\leq t}) = \text{softmax}(W_k^{\text{exit}} \cdot \text{LN}(\bar{\mathbf{h}}_t^{(k)}) + \mathbf{b}_k^{\text{exit}}), \quad (21)$$

where $\bar{\mathbf{h}}_t^{(k)}$ is the merged representation at the end of stage k (or the mean over streams if exit is evaluated before merge).

3.5.1 Cross-Stream Agreement as Confidence

When multiple streams are active, we define a *cross-stream agreement* score:

$$\text{Agree}_t^{(k)} = \frac{1}{\binom{S_k}{2}} \sum_{s < s'} \cos(\mathbf{h}_{t,s}^{(k,L_k)}, \mathbf{h}_{t,s'}^{(k,L_k)}), \quad (22)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity. This measures the degree to which independent processing streams have converged to the same representation for token t .

Definition 3.7 (Early exit criterion). Token t exits at stage k if:

$$\text{Agree}_t^{(k)} > \tau_{\text{conf}} \quad \text{and} \quad H[\hat{p}_k(\cdot \mid \mathbf{x}_{\leq t})] < \tau_{\text{ent}}, \quad (23)$$

where $H[\cdot]$ denotes entropy and $\tau_{\text{conf}}, \tau_{\text{ent}}$ are thresholds. The first condition requires cross-stream consensus (Felix’s convergence detection); the second requires the prediction itself to be low-entropy.

3.5.2 Training Objective with Deep Supervision

During training, all tokens traverse all stages (no early exit). The loss combines predictions from all exit points and the final output:

$$\mathcal{L} = \sum_{k=0}^{K-1} \lambda_k \cdot \mathcal{L}_k, \quad \text{where} \quad \mathcal{L}_k = -\frac{1}{T} \sum_{t=1}^T \log \hat{p}_k(x_{t+1} \mid \mathbf{x}_{\leq t}), \quad (24)$$

with weights λ_k satisfying $\sum_k \lambda_k = 1$ and $\lambda_{K-1} > \lambda_k$ for $k < K - 1$ (the final stage is weighted most heavily). This deep supervision encourages useful representations at every stage, enabling meaningful early exit at inference time.

3.6 Output Head

The final output is produced from the single stream at the end of stage $K - 1$:

$$p(x_{t+1} \mid \mathbf{x}_{\leq t}) = \text{softmax}(W^{\text{out}} \cdot \text{LN}(\mathbf{h}_{t,1}^{(K-1,L_{K-1})}) + \mathbf{b}^{\text{out}}), \quad (25)$$

where $W^{\text{out}} \in \mathbb{R}^{V \times d_{K-1}}$ may optionally be tied to the input embedding matrix (transposed).

3.7 Full Forward Pass

For completeness, the full forward pass for a sequence $\mathbf{x}$ is given in Algorithm 1.

Algorithm 1 Felix-LM Forward Pass

Require: Sequence $\mathbf{x} = (x_1, \dots, x_T)$; stages $k = 0, \dots, K-1$

- 1: $\mathbf{e}_t \leftarrow \text{Embed}(x_t)$ for all t
- 2: **for** $s = 1, \dots, S_0$ **do** ▷ Initialize streams
- 3: $\mathbf{h}_{t,s}^{(0,0)} \leftarrow W_s^{\text{init}} \mathbf{e}_t + \mathbf{b}_s^{\text{init}}$ for all t
- 4: **end for**
- 5: **for** $k = 0, \dots, K-1$ **do** ▷ Process stages
- 6: **for** $s = 1, \dots, S_k$ **do** ▷ Independent streams
- 7: **for** $\ell = 0, \dots, L_k - 1$ **do** ▷ Layers within stream
- 8: $\mathbf{h}_{:,s}^{(k,\ell+1)} \leftarrow \text{TransformerLayer}_{k,\ell,s}(\mathbf{h}_{:,s}^{(k,\ell)})$ ▷ Eq. 10–11
- 9: **end for**
- 10: **end for**
- 11: **if** $k < K-1$ **then** ▷ Merge into next stage
- 12: Pair streams: $(s_1, s_2), (s_3, s_4), \dots$
- 13: **for** each pair (s, s') **do**
- 14: $\mathbf{h}_{:, \text{new}}^{(k+1,0)} \leftarrow \text{Merge}_k(\mathbf{h}_{:,s}^{(k,L_k)}, \mathbf{h}_{:,s'}^{(k,L_k)})$ ▷ Eq. 14
- 15: **end for**
- 16: Compute exit loss $\mathcal{L}_k$ ▷ Eq. 24
- 17: **end if**
- 18: **end for**
- 19: **return** $\text{softmax}(W^{\text{out}} \cdot \text{LN}(\mathbf{h}_{:,1}^{(K-1,L_{K-1})}))$

4 Theoretical Analysis

4.1 Information-Theoretic Capacity

We analyze the representational capacity of the multi-stream architecture relative to a uniform transformer of equal parameter count.

Proposition 4.1 (Parameter distribution). *Consider a Felix-LM model with K stages, stream counts $(S_0, \dots, S_{K-1})$, dimensions $(d_0, \dots, d_{K-1})$, and layer counts $(L_0, \dots, L_{K-1})$. Ignoring embedding and output layers, the total parameter count is approximately:*

$$N_{\text{Felix}} \approx \sum_{k=0}^{K-1} S_k \cdot L_k \cdot (4d_k^2 + 8d_k^2) + \sum_{k=0}^{K-2} (4d_k \cdot d_{k+1} + 8d_k^2), \quad (26)$$

where $4d_k^2$ accounts for attention projections (W_Q, W_K, W_V, W_O , each $d_k \times d_k$, simplified for single-head), $8d_k^2$ for the FFN (two layers of $d_k \times 4d_k$), and the merge terms account for the gating and projection matrices.

A uniform transformer with L layers of dimension d has:

$$N_{\text{uniform}} \approx L \cdot 12d^2. \quad (27)$$

The design constraint is $N_{\text{Felix}} \approx N_{\text{uniform}}$ for fair comparison. Given this constraint, the Felix-LM model distributes parameters differently: more parameters are allocated to early-stage parallel streams (broad exploration) and fewer to late-stage computation (focused synthesis, but only one stream).

Proposition 4.2 (Representation diversity bound). *At the output of Stage 0, the S_0 independent streams produce S_0 distinct representations for each token. Under the INDEPENDENT weight configuration, the total representational capacity (measured as the dimension of the space of possible token representations) is:*

$$\dim(\text{range of Stage 0}) = S_0 \cdot d_0, \quad (28)$$

which exceeds the capacity d_0 of any individual stream. The merge operation projects this $S_0 \cdot d_0$ -dimensional representation down to d_1 -dimensional, with the gate $\mathbf{g}_k$ performing learned dimensionality selection.

For a uniform transformer, the representation at any intermediate layer has capacity d_{model} . If $S_0 \cdot d_0 > d_{\text{model}}$ (which holds whenever the streams collectively span more dimensions than the uniform model’s width), Stage 0 of Felix-LM can represent a strictly richer space of initial features before compression.

4.2 Gradient Flow Analysis

A critical concern with dimension-changing architectures is gradient flow degradation. We analyze the gradient of the loss with respect to early-stage parameters.

Proposition 4.3 (Gradient path through merges). *Let $\mathcal{L}$ be the training loss and consider the gradient with respect to a parameter θ in stream s of Stage 0. The gradient flows backward through:*

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(K-1)}} \cdot \prod_{k=K-2}^0 \underbrace{\frac{\partial \mathbf{h}^{(k+1)}}{\partial \mathbf{h}^{(k)}}}_{\text{merge Jacobian}} \cdot \frac{\partial \mathbf{h}^{(0)}}{\partial \theta}. \quad (29)$$

The merge Jacobian has the form:

$$J_k^{\text{merge}} = P_k \cdot \text{diag}(\mathbf{g}_k + \mathbf{g}'_k \odot [\mathbf{h}_s; \mathbf{h}_{s'}]), \quad (30)$$

where $\mathbf{g}'_k = \mathbf{g}_k \odot (1 - \mathbf{g}_k)$ is the sigmoid derivative. If the gate saturates ($g_{k,i} \rightarrow 0$ or $g_{k,i} \rightarrow 1$), the Jacobian for that dimension reduces to $P_k \cdot \text{diag}(\mathbf{g}_k)$ (the derivative term vanishes), and gradient flows through the non-gated pathway.

Corollary 4.4 (Gradient preservation under orthogonal merges). *If P_k has orthonormal rows ($P_k P_k^\top = I$), then the merge preserves gradient norms in the following sense:*

$$\|J_k^{\text{merge}} \mathbf{v}\| \geq \min_i(g_{k,i}) \cdot \|\mathbf{v}\| \quad (31)$$

for any gradient vector $\mathbf{v}$, provided all gates are open ($g_{k,i} = 1$). Gradient vanishing through the merge is thus controlled by the gate values, not by the projection geometry.

This motivates initializing the gate biases $\mathbf{b}_k^g$ to positive values (gates start open) and optionally constraining P_k to be orthogonal, especially in deeper models.

Deep supervision as gradient highway. The deep supervision loss (eq. (24)) provides an additional gradient pathway: each exit head injects gradients directly into its corresponding stage without passing through downstream merges. This ensures that even if merge gradients attenuate, every stage receives direct supervisory signal.

4.3 Representational Geometry

We now establish that the Felix-LM architecture induces the helical funnel trajectory of definition 2.3.

Theorem 4.5 (Helical funnel trajectory). *Under the Felix-LM architecture with depth-extended RoPE (definition 3.4) and gated merge operations (definition 3.2), the representation trajectory $\gamma(\mathbf{x})$ satisfies definition 2.3 with:*

- (a) Angular velocity determined by the depth-RoPE frequencies θ_i^{depth} within stages and by the polar decomposition of P_k at merge boundaries;

- (b) Radial compression rate $\rho^{(\ell)} = 1$ within stages (constant dimensionality) and $\rho^{(k \rightarrow k+1)} = d_{k+1}/(S_k \cdot d_k)$ at merges;
- (c) Effective dimensionality decreasing monotonically as $d_{\text{eff}}^{(\text{after stage } k)} = d_{k+1}$, from an initial capacity of $S_0 \cdot d_0$ to a final capacity of d_{K-1} .

Proof sketch. Within a stage, RMSNorm preserves the subspace, and depth-extended RoPE applies an explicit rotation via the depth-frequency term in eq. (19). This satisfies the $Q^{(\ell)}$ term of eq. (5) with the gate being the identity ($\mathbf{g} = \mathbf{1}$, no compression within stages).

At a merge boundary, the concatenation of S_k stream representations embeds them in $\mathbb{R}^{S_k \cdot d_k}$. The gated projection $P_k(\mathbf{g}_k \odot [\cdot])$ decomposes via polar decomposition (eq. (16)) into rotation $(V_k^\top, U_k)$ and scaling (Σ_k) , with the gate $\mathbf{g}_k$ providing additional soft dimensionality reduction. The post-merge dimension $d_{k+1} < S_k \cdot d_k$ gives $\rho < 1$, completing the compression step.

The trajectory therefore alternates between rotation-only phases (within-stage, via depth-RoPE) and rotation-with-compression phases (at merges), tracing a helical funnel as defined. $\square$

5 Connection to the Felix Framework

The architecture is parameterized to allow direct configuration from the Felix framework’s YAML specification. The mapping is:

Felix Parameter	Felix-LM Analog	Mapping
top_radius (R_{max})	Initial stream count S_0	$S_0 = 2^{\lceil \log_2 R_{\text{max}} \rceil}$
bottom_radius (R_{min})	Final stream count S_{K-1}	$S_{K-1} = 1$ (always)
turns (n)	Stage transitions $K - 1$	$K = n + 1$
height (H)	Total layers L_{total}	$L_{\text{total}} \propto H$
confidence_threshold	Exit threshold τ_{conf}	Direct mapping
temperature_range	Attention type per stage	See §3.2
compression_ratio	Merge compression	$d_{k+1}/(2d_k)$

The exponential tapering function (eq. (4)) determines the rate of stream merging: if $R(t)$ decreases exponentially, stream counts decrease by factors of 2 at each stage (binary merging), yielding $S_k = S_0/2^k$. Alternative merging schedules (e.g., merging 3-to-1) correspond to steeper radius decay.

6 Proposed Experiments

6.1 Ablation Study Design

All models are trained on WikiText-103 [7] with identical optimization hyperparameters. The evaluation metric is perplexity on the held-out test set.

Model	Name	Configuration
M0	UNIFORM	Standard transformer, L layers, d width. Parameter-matched baseline.
M1	MSPM-HOMO	Multi-stream progressive merging; all stages use standard causal attention. Tests the stream-merge structure alone.
M2	MSPM-HETERO	As M1, but Stage 0 uses linear attention, intermediate stages use sliding-window, final stage uses full causal attention. Tests heterogeneous computation.
M3	MSPM-EXIT	As M2, plus early-exit heads with cross-stream agreement gating. Tests adaptive depth.
M4	NARROW- d	Single stream throughout, but d_{model} decreases across stages (the “narrow funnel” hypothesis). Controls for stream merging vs. dimension narrowing.

Controlled variables. Total parameter count $N \approx 10\text{M}$ ($\pm 5\%$), context length $T = 512$, vocabulary from WikiText-103 BPE tokenizer, AdamW optimizer with cosine learning rate schedule, batch size, warmup steps, and total training tokens are held constant across all models.

6.2 Hypotheses

- H1.** M1 achieves lower perplexity than M0, indicating that the multi-stream progressive merging structure provides a useful inductive bias for language modeling.
- H2.** M2 achieves lower perplexity than M1, indicating that heterogeneous attention types aligned to stage roles improve performance beyond the structural bias alone.
- H3.** M3 achieves comparable or better perplexity to M2 while reducing average inference FLOPs (via early exit on easy tokens).
- H4.** M4 achieves worse perplexity than M1, indicating that the funnel should operate on stream count rather than model dimension.

6.3 Diagnostic Metrics

Beyond perplexity, we propose monitoring:

- **Cross-stream agreement** (eq. (22)) over training—do streams converge to specialization or redundancy?
- **Gate activation statistics:** distribution of $\mathbf{g}_k$ values at each merge, tracking which dimensions are preserved versus suppressed.
- **Exit rate by stage:** at inference time, what fraction of tokens exit at each stage?
- **Per-stage loss:** $\mathcal{L}_k$ for each stage, measuring whether early stages learn useful intermediate predictions.
- **Effective dimensionality** (definition 2.2) at each merge output, measuring realized compression.

7 Extensions and Open Questions

Sequence-dimension funneling. The current design funnels along the stream/feature axis. An orthogonal extension is to pool tokens at merge boundaries (reducing sequence length), providing a second axis of compression. The interaction between feature-funneling and sequence-funneling is unexplored and theoretically interesting: it corresponds to the Felix helix narrowing in both radius (feature) and height per turn (sequence).

Skip connections across stages. The depth-extended RoPE provides a natural framework for cross-stage residual connections, since representations at different depths are in well-defined rotational relationships. A skip connection from Stage 0 to Stage 2 would require projecting across different dimensions, but the rotary structure provides a principled way to align the coordinate frames.

Scaling laws. If the architecture proves competitive at 10M parameters, the next question is whether the multi-stream structure obeys favorable scaling laws. The additional parameters in merge operations and multiple stream initializations represent overhead; whether this overhead amortizes favorably with scale is an empirical question.

Hybrid SSM-Attention. The heterogeneous attention strategy (§3.2) points toward a principled framework for combining state-space models (Mamba, RWKV) with attention: use SSMs for cheap broad context gathering in early stages and attention for precise synthesis in late stages. This is consonant with the Felix principle that early agents are broad/cheap and late agents are focused/expensive.

Non-binary merging. The current design merges streams in pairs. For S_0 that is not a power of 2, or for more aggressive funneling, k -way merges (concatenating $k > 2$ streams before projection) may be preferable. The gating and projection mathematics extend naturally.

8 Related Work

Funnel architectures. FunnelTransformer [8] narrows the sequence dimension via pooling in encoder models. Felix-LM differs in two ways: it narrows the stream/feature axis (not sequence), and it targets autoregressive (causal) language modeling rather than bidirectional encoding.

Mixture of Experts / Mixture of Depths. Mixture-of-Experts models [9, 10] route tokens to different sub-networks at each layer. Felix-LM’s multi-stream structure is superficially similar but structurally distinct: in MoE, routing is per-token and per-layer; in Felix-LM, all tokens follow the same multi-stream structure, and the streams interact only at merge points. Mixture of Depths [11] allows tokens to skip layers, which is related to our early-exit mechanism.

Rotary embeddings. RoPE [5] and its extensions (YaRN [12], NTK-aware scaling) manipulate the rotation frequencies for context extension. Our depth-extended RoPE adds a new axis (layer index) to the rotation, which to our knowledge has not been explored in the literature.

Adaptive computation. Universal Transformer [13] and PonderNet [14] implement depth-adaptive computation via halting mechanisms. Felix-LM’s cross-stream agreement provides a novel confidence signal unavailable in single-stream architectures.

Geometric deep learning. Clifford Algebra Networks [15] and related work on rotation-equivariant architectures [16, 17] provide rigorous frameworks for incorporating geometric structure. Felix-LM’s helical trajectory is a softer form of geometric inductive bias: it does not enforce strict equivariance but encourages helical representation dynamics through architectural structure and initialization.

9 Discussion

The central hypothesis of this work is that the principles underlying the Felix multi-agent framework—parallel exploration, progressive convergence, depth-dependent computation, and confidence-gated resource allocation—provide a useful inductive bias when instantiated in the architecture of a causal language model.

We have been deliberately conservative in our claims. The proposed architecture introduces complexity: multiple streams increase memory requirements (though not FLOPs, since streams can be processed sequentially), merge operations add parameters, and heterogeneous attention types complicate implementation. Whether the inductive bias is strong enough to compensate for this added complexity is an empirical question that the proposed ablation study is designed to answer.

The most novel contribution, independent of whether the full architecture proves competitive, is the *cross-stream agreement* signal for confidence estimation. This provides a measure of representational robustness—the degree to which independent processing pathways converge on the same answer—that has no analog in single-stream architectures and may find applications in uncertainty quantification and hallucination detection beyond language modeling.

Acknowledgments

The Felix multi-agent framework was developed by Caleb [Surname]. The Multi-Stream Progressive Merging architecture was developed through iterative collaboration between C. [Surname] and AI design assistants (Claude, Anthropic). The authors thank [physicist and mathematician reviewers] for their feedback on the mathematical formalism.

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A Initialization Strategy

The stream initialization projections W_s^{init} (eq. (9)) should encourage diverse initial representations. We propose initializing them as orthogonal subspace projections.

Let $d_{\text{embed}} \geq S_0 \cdot d_0$. Generate a random orthogonal matrix $Q \in \mathbb{R}^{d_{\text{embed}} \times d_{\text{embed}}}$ (via QR decomposition of a random Gaussian matrix) and partition its rows into S_0 blocks of size d_0 :

$$W_s^{\text{init}} = Q_{[(s-1)d_0:s \cdot d_0, :]}. \quad (32)$$

This ensures that the initial stream representations are projections onto orthogonal subspaces of the embedding, maximizing initial diversity. Under this initialization:

$$\langle W_s^{\text{init}} \mathbf{e}, W_{s'}^{\text{init}} \mathbf{e} \rangle = 0 \quad \text{for } s \neq s', \quad (33)$$

for any input $\mathbf{e}$, guaranteeing that streams start with completely independent views of the input.

B Parameter Count Calculations

We provide exact parameter count formulas for the ablation models, parameterized to target $N \approx 10\text{M}$.

For each model, the total parameters are:

$$N = N_{\text{embed}} + N_{\text{stages}} + N_{\text{merges}} + N_{\text{exits}} + N_{\text{output}},$$

where:

$$N_{\text{embed}} = V \cdot d_{\text{embed}} + S_0 \cdot d_0 \cdot d_{\text{embed}}, \quad (34)$$

$$N_{\text{stages}} = \sum_{k=0}^{K-1} S_k \cdot L_k \cdot (4d_k^2 + 2 \cdot d_k \cdot 4d_k), \quad (35)$$

$$N_{\text{merges}} = \sum_{k=0}^{K-2} \frac{S_k}{2} \cdot (d_{k+1} \cdot 2d_k + 2d_k \cdot 2d_k), \quad (36)$$

$$N_{\text{exits}} = \sum_{k=0}^{K-2} (d_k \cdot V), \quad (37)$$

$$N_{\text{output}} = d_{K-1} \cdot V. \quad (38)$$

Concrete configurations for each model will be computed numerically under the constraint $N = 10 \times 10^6 \pm 5\%$.